SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Code of Conduct

I M. Greeshma / 192521195 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Greeshma

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

Definition :- cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet, allowing users to access resources whenever required without owing physical infrastructure.

four characteristics :-

i) on-demand self-service: users can access computing resources whenever needed without direct assistence from the service provider.

ii) Broad network access :- cloud services can be accessed through the internet using computers, laptops, tablets and smartphones

iii) Resource pooling :- The provider shares computing requires among multiple users according to their needs.

iv) Rapid electricity :- Resources can be increased or decreased quickly based on demand.

Difference :- Traditional on-premise, computing requires organizations to purchase, maintain, and manage their own hardware, whereas cloud computing. Provides flexible resources over the internet.

2) Cloud computing envolved through several techn developments :

1) main frame computing :- centralized computers multiple users through terminals

2) Personal computers :- made computing affordable assessable

3) client - server computing :- clients acessed from networked servers.

4) virtualization :- multiple virtual machines on one physical server.

5) Internet development :- Enabled remote access to computing resources.

6) Web Application and SOA :- Enabled online softwa and communication b/w applications

7) Modern cloud platforms :- Automation, containis, and distributed storage enabled scalable cloud services like AWS, Azure

conclusion :- Hardware evolution, virtualization, the internet, and web technologies led to modern cloud computing.

Server virtualization in cloud computing

→ divides one physical server into multiple virtual machines using a hypervisor

Roles:-

→ improves hardware utilization

→ reduces costs

→ enables flexible resource allocation

→ supports quick VM creation

→ provides workload isolation

Virtualization	cloud computing
Technology that creates virtual resources	Delivers computing resources over a network
Can operate on local servers	provides on-demand scalable services
Eg:- Exi, VMware	Ex:- AWS

conclusion:- virtualization is an enabling technology, while cloud computing delivers services using technologies such as virtualization.

4) Roles of Data in cloud computing

→ A data center is a facility containing servers, storage, networking, power, and cooling systems

Roles :-

1) storage : stores files and databases

2) processing : Runs application

3) networking : connects cloud resources to users

4) Availability : provides backup and recovery

5) security : protects data and infrastructure

6) scalability : Expands resources according to demand.

Ex :- AWS data centers support cloud servers and storage

Conclusion :- Data centers provide the physical infrastructure for reliable and scalable cloud service.

Paas, Saas and Xaas

	full form	description	Example
Iaas	infrastructure as a service	provides virtual serveces & storage	Amazon EC2
Paas	platform as a service	Provides an application development Platform	Google App Engine
Saas	Software as a service	provides ready-to use online softwar	Google Docs
Xaas	Anything as a service	delivers various IT resources as services	Dropbox (cloud storage).

Key differences :-

- Iaas :- users manage the OS and applications.
- Paas :- users develop and manage applications.
- Saas :- users simply use the software
- Xaas :- covers many types of services delivered online

Conclusion :- these models provide different levels of control and convenience based on users needs.

10/10/26